

System of Linear Equations



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Existence and uniqueness of solutions

Introduction: Existence and uniqueness of solutions

- ▶ A system of linear equations may have no solution, exactly one solution or infinitely many solutions
- ▶ In this section we see how to decide which case occurs, **before** solving the system, by comparing two ranks: the rank of the coefficient matrix and the rank of the augmented matrix
- ▶ This is the content of the **Rouché–Capelli theorem**, which we will use throughout the chapter

System of linear equations

Consider a system of m linear equations with n variables:

$$\mathbf{A} \mathbf{x} = \mathbf{b}$$

where:

- ▶ $\mathbf{A} \in \mathbb{R}^{m \times n}$ is the coefficient matrix (m equations, n variables)
- ▶ $\mathbf{x} \in \mathbb{R}^{n \times 1}$ is the vector of variables
- ▶ $\mathbf{b} \in \mathbb{R}^{m \times 1}$ is the right-hand side vector
- ▶ $(\mathbf{A}|\mathbf{b}) \in \mathbb{R}^{m \times (n+1)}$ is the augmented matrix

Rouché–Capelli theorem (1/2)

The existence and uniqueness of solutions to the system $\mathbf{A} \mathbf{x} = \mathbf{b}$ depend on the rank of the coefficient matrix $\mathbf{A}$ and the rank of the augmented matrix $(\mathbf{A}|\mathbf{b})$.

Case 1 - No solution (inconsistent system):

$$\text{rank}(\mathbf{A}) < \text{rank}(\mathbf{A}|\mathbf{b})$$

The system is inconsistent and has no solution.

Rouché–Capelli theorem (2/2)

Case 2 - Unique solution:

$$\text{rank}(\mathbf{A}) = \text{rank}(\mathbf{A}|\mathbf{b}) = n$$

The system has a unique solution.

Case 3 - Infinite solutions:

$$\text{rank}(\mathbf{A}) = \text{rank}(\mathbf{A}|\mathbf{b}) < n$$

The system has infinitely many solutions with $n - \text{rank}(\mathbf{A})$ degrees of freedom (free variables). If $r = \text{rank}(\mathbf{A})$, the solutions can be described by means of $n - r$ **free parameters**.

Rouché–Capelli theorem: remarks (1/2)

- ▶ Since $(\mathbf{A}|\mathbf{b})$ is obtained from $\mathbf{A}$ by adding one column, we always have

$$\text{rank}(\mathbf{A}) \leq \text{rank}(\mathbf{A}|\mathbf{b}) \leq \text{rank}(\mathbf{A}) + 1,$$

so the three cases above cover all the possibilities.

- ▶ In particular, the system has at least one solution (it is **consistent**) if and only if $\text{rank}(\mathbf{A}) = \text{rank}(\mathbf{A}|\mathbf{b})$.

Rouché–Capelli theorem: remarks (2/2)

- ▶ Since $\text{rank}(\mathbf{A}) \leq \min\{m, n\}$, a unique solution is possible only if $m \geq n$.
- ▶ If $\mathbf{A} \in \mathbb{R}^{n \times n}$ is square, recall that $\text{rank}(\mathbf{A}) = n$ if and only if $\det(\mathbf{A}) \neq 0$.
- ▶ Hence, if $\det(\mathbf{A}) \neq 0$, then also $\text{rank}(\mathbf{A}|\mathbf{b}) = n$ and the system has a unique solution, for every right-hand side $\mathbf{b}$.

Summary

Rouché–Capelli theorem for $\mathbf{A} \mathbf{x} = \mathbf{b}$, $\mathbf{A} \in \mathbb{R}^{m \times n}$, $r = \text{rank}(\mathbf{A})$:

- ▶ $\text{rank}(\mathbf{A}) < \text{rank}(\mathbf{A}|\mathbf{b})$: no solution
- ▶ $\text{rank}(\mathbf{A}) = \text{rank}(\mathbf{A}|\mathbf{b}) = n$: unique solution
- ▶ $\text{rank}(\mathbf{A}) = \text{rank}(\mathbf{A}|\mathbf{b}) < n$: infinitely many solutions, with $n - r$ free parameters

Square systems:

- ▶ If $\det(\mathbf{A}) \neq 0$, the solution is unique for every $\mathbf{b}$

Gaussian Elimination Method



Introduction: Gaussian Elimination Method

- ▶ Gaussian elimination is the standard algorithm to solve a system of linear equations by hand and on a computer
- ▶ The idea is to transform the system into an equivalent **triangular** system, which is then easy to solve from the bottom to the top
- ▶ We will see the elementary row operations, the two phases of the method and a complete example

Gaussian elimination

The **Gaussian elimination method** is an algorithm for solving systems of linear equations by transforming the augmented matrix $(\mathbf{A}|\mathbf{b})$ into **row echelon form** (upper triangular form) through elementary row operations.

The two phases of the method

- ▶ Given the system of linear equations:

$$\mathbf{A} \mathbf{x} = \mathbf{b}$$

we form the **augmented matrix**:

$$(\mathbf{A}|\mathbf{b})$$

- ▶ The method consists of two phases:
 - ▶ **Phase 1 - Forward elimination**
 - ▶ **Phase 2 - Backward substitution**

Phase 1 - Forward elimination

Transform the augmented matrix into upper triangular form using elementary row operations:

- ▶ Swap two rows
- ▶ Multiply a row by a non-zero scalar
- ▶ Add a multiple of one row to another row

The goal is to create zeros below the diagonal, obtaining:

$$\left(\begin{array}{cccc|c} \tilde{a}_{11} & \tilde{a}_{12} & \cdots & \tilde{a}_{1n} & \tilde{b}_1 \\ 0 & \tilde{a}_{22} & \cdots & \tilde{a}_{2n} & \tilde{b}_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & \tilde{a}_{nn} & \tilde{b}_n \end{array} \right)$$

Phase 2 - Backward substitution

Solve the upper triangular system from bottom to top (assuming $\tilde{a}_{ii} \neq 0$ for all i):

$$\begin{aligned}x_n &= \frac{\tilde{b}_n}{\tilde{a}_{nn}} \\x_{n-1} &= \frac{\tilde{b}_{n-1} - \tilde{a}_{n-1,n} x_n}{\tilde{a}_{n-1,n-1}} \\&\vdots \\x_i &= \frac{\tilde{b}_i - \sum_{j=i+1}^n \tilde{a}_{ij} x_j}{\tilde{a}_{ii}}\end{aligned}$$

Elementary row operations

- ▶ $R_i \leftrightarrow R_j$: swap row i with row j
- ▶ $R_i \leftarrow k R_i$: multiply row i by scalar $k \neq 0$
- ▶ $R_i \leftarrow R_i + k R_j$: add k times row j to row i

These operations do not change the solution set of the system.

Example: Gaussian elimination (1/5)

Consider the system of linear equations:

$$\begin{cases} 2x_1 + 1x_2 + 1x_3 = 4 \\ 4x_1 + 3x_2 + 3x_3 = 10 \\ 8x_1 + 7x_2 + 9x_3 = 24 \end{cases}$$

Augmented matrix:

$$(\mathbf{A}|\mathbf{b}) = \left(\begin{array}{ccc|c} 2 & 1 & 1 & 4 \\ 4 & 3 & 3 & 10 \\ 8 & 7 & 9 & 24 \end{array} \right)$$

Example: Gaussian elimination (2/5)

Step 1: Eliminate x_1 from rows 2 and 3

$R_2 \leftarrow R_2 - 2 R_1$ (subtract 2 times row 1 from row 2):

$$\left(\begin{array}{ccc|c} 2 & 1 & 1 & 4 \\ 0 & 1 & 1 & 2 \\ 8 & 7 & 9 & 24 \end{array} \right)$$

$R_3 \leftarrow R_3 - 4 R_1$ (subtract 4 times row 1 from row 3):

$$\left(\begin{array}{ccc|c} 2 & 1 & 1 & 4 \\ 0 & 1 & 1 & 2 \\ 0 & 3 & 5 & 8 \end{array} \right)$$

Example: Gaussian elimination (3/5)

Step 2: Eliminate x_2 from row 3

$R_3 \leftarrow R_3 - 3 R_2$ (subtract 3 times row 2 from row 3):

$$\left(\begin{array}{ccc|c} 2 & 1 & 1 & 4 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 2 & 2 \end{array} \right)$$

Upper triangular form achieved! The corresponding system is:

$$\begin{cases} 2x_1 + 1x_2 + 1x_3 & = 4 \\ 1x_2 + 1x_3 & = 2 \\ 2x_3 & = 2 \end{cases}$$

Example: Gaussian elimination (4/5)

Backward substitution:

From the third equation:

$$2 x_3 = 2 \implies x_3 = 1$$

From the second equation:

$$x_2 + x_3 = 2 \implies x_2 = 2 - x_3 = 2 - 1 = 1$$

From the first equation:

$$\begin{aligned} 2 x_1 + x_2 + x_3 &= 4 \implies 2 x_1 = 4 - x_2 - x_3 = 4 - 1 - 1 = 2 \\ \implies x_1 &= 1 \end{aligned}$$

Example: Gaussian elimination (5/5)

Solution:

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

Verification:

$$\mathbf{A} \mathbf{x} = \begin{pmatrix} 2 & 1 & 1 \\ 4 & 3 & 3 \\ 8 & 7 & 9 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 2+1+1 \\ 4+3+3 \\ 8+7+9 \end{pmatrix} = \begin{pmatrix} 4 \\ 10 \\ 24 \end{pmatrix} = \mathbf{b} \quad \checkmark$$

Summary

Gaussian elimination:

- ▶ Work on the augmented matrix $(\mathbf{A}|\mathbf{b})$
- ▶ Phase 1: forward elimination with elementary row operations, to obtain zeros below the diagonal
- ▶ Phase 2: backward substitution, from the last equation to the first one

Elementary row operations:

- ▶ $R_i \leftrightarrow R_j, \quad R_i \leftarrow k R_i \ (k \neq 0), \quad R_i \leftarrow R_i + k R_j$
- ▶ They do not change the solution set of the system

Gaussian elimination and the Rouché–Capelli theorem



Introduction: Gaussian elimination and the Rouché–Capelli theorem

- ▶ Gaussian elimination works for **any** system, also when the solution is not unique or does not exist
- ▶ At the end of forward elimination we can read the two ranks and apply the Rouché–Capelli theorem
- ▶ We will see three worked examples: a unique solution, infinitely many solutions (in parametric form) and no solution

Row echelon form (1/2)

- ▶ Gaussian elimination can be applied to any system $\mathbf{A} \mathbf{x} = \mathbf{b}$, with $\mathbf{A} \in \mathbb{R}^{m \times n}$.
- ▶ In general, forward elimination transforms $(\mathbf{A}|\mathbf{b})$ into **row echelon form**: in each nonzero row, the first nonzero entry (called **pivot**) is strictly to the right of the pivot of the row above, and the rows of zeros (if any) are at the bottom.
- ▶ Elementary row operations do not change the solution set of the system, and they do not change the rank of $\mathbf{A}$ nor the rank of $(\mathbf{A}|\mathbf{b})$.

Row echelon form (2/2)

- ▶ The rank of a matrix in row echelon form is equal to the number of its nonzero rows.
- ▶ Therefore, at the end of forward elimination we can read $\text{rank}(\mathbf{A})$ and $\text{rank}(\mathbf{A}|\mathbf{b})$ and apply the Rouché–Capelli theorem:
 - ▶ a row of the form $(0 \ \cdots \ 0 \mid c)$ with $c \neq 0$ corresponds to the impossible equation $0 = c$: then $\text{rank}(\mathbf{A}) < \text{rank}(\mathbf{A}|\mathbf{b})$ and the system has no solution;
 - ▶ otherwise, the variables whose columns contain a pivot are computed by backward substitution, and the remaining $n - r$ variables are the **free parameters**.

Parametric form

When the system has infinitely many solutions with one free parameter $t \in \mathbb{R}$, the solutions can be written in **parametric form**

$$\mathbf{x} = \mathbf{x}_0 + t \mathbf{v}, \quad t \in \mathbb{R},$$

where $\mathbf{x}_0$ is a particular solution (obtained for $t = 0$) and $\mathbf{v}$ is a nonzero vector such that $\mathbf{A} \mathbf{v} = \mathbf{0}$.

Example: unique solution (1/3)

Consider the system of linear equations:

$$\begin{cases} 1 x_2 + 1 x_3 = 1 \\ 1 x_1 + 1 x_2 + 1 x_3 = 2 \\ 2 x_1 + 1 x_2 + 3 x_3 = 1 \end{cases} \quad (\mathbf{A}|\mathbf{b}) = \left(\begin{array}{ccc|c} 0 & 1 & 1 & 1 \\ 1 & 1 & 1 & 2 \\ 2 & 1 & 3 & 1 \end{array} \right)$$

The entry in position (1, 1) is zero, so we first swap rows 1 and 2.

$R_1 \leftrightarrow R_2$:

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 0 & 1 & 1 & 1 \\ 2 & 1 & 3 & 1 \end{array} \right)$$

Example: unique solution (2/3)

$$R_3 \leftarrow R_3 - 2 R_1:$$

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 0 & 1 & 1 & 1 \\ 0 & -1 & 1 & -3 \end{array} \right)$$

$$R_3 \leftarrow R_3 + R_2:$$

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 2 & -2 \end{array} \right)$$

Example: unique solution (3/3)

There are 3 nonzero rows both in $\mathbf{A}$ and in $(\mathbf{A}|\mathbf{b})$:

$$\text{rank}(\mathbf{A}) = \text{rank}(\mathbf{A}|\mathbf{b}) = 3 = n,$$

hence the solution is unique. By backward substitution:

$$2x_3 = -2 \implies x_3 = -1, \quad x_2 = 1 - x_3 = 2,$$

$$x_1 = 2 - x_2 - x_3 = 2 - 2 + 1 = 1,$$

that is,

$$\mathbf{x} = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}.$$

Example: infinitely many solutions (1/4)

Consider the system of linear equations:

$$\begin{cases} 1 x_1 + 2 x_2 - 1 x_3 = 1 \\ 2 x_1 + 5 x_2 + 1 x_3 = 4 \\ 3 x_1 + 7 x_2 = 5 \end{cases} \quad (\mathbf{A}|\mathbf{b}) = \left(\begin{array}{ccc|c} 1 & 2 & -1 & 1 \\ 2 & 5 & 1 & 4 \\ 3 & 7 & 0 & 5 \end{array} \right)$$

$$R_2 \leftarrow R_2 - 2 R_1:$$

$$\left(\begin{array}{ccc|c} 1 & 2 & -1 & 1 \\ 0 & 1 & 3 & 2 \\ 3 & 7 & 0 & 5 \end{array} \right)$$

Example: infinitely many solutions (2/4)

$$R_3 \leftarrow R_3 - 3 R_1:$$

$$\left(\begin{array}{ccc|c} 1 & 2 & -1 & 1 \\ 0 & 1 & 3 & 2 \\ 0 & 1 & 3 & 2 \end{array} \right)$$

$$R_3 \leftarrow R_3 - R_2:$$

$$\left(\begin{array}{ccc|c} 1 & 2 & -1 & 1 \\ 0 & 1 & 3 & 2 \\ 0 & 0 & 0 & 0 \end{array} \right)$$

Example: infinitely many solutions (3/4)

There are 2 nonzero rows both in $\mathbf{A}$ and in $(\mathbf{A}|\mathbf{b})$:

$$\text{rank}(\mathbf{A}) = \text{rank}(\mathbf{A}|\mathbf{b}) = 2 < 3 = n,$$

hence there are infinitely many solutions with $n - r = 3 - 2 = 1$ free parameter.

The pivots are in the columns of x_1 and x_2 , so x_3 is free: we set $x_3 = t$, $t \in \mathbb{R}$. By backward substitution:

$$x_2 = 2 - 3x_3 = 2 - 3t,$$

$$x_1 = 1 - 2x_2 + x_3 = 1 - 4 + 6t + t = -3 + 7t.$$

Example: infinitely many solutions (4/4)

In parametric form:

$$\mathbf{x} = \begin{pmatrix} -3 + 7t \\ 2 - 3t \\ t \end{pmatrix} = \underbrace{\begin{pmatrix} -3 \\ 2 \\ 0 \end{pmatrix}}_{\mathbf{x}_0} + t \underbrace{\begin{pmatrix} 7 \\ -3 \\ 1 \end{pmatrix}}_{\mathbf{v}}, \quad t \in \mathbb{R}.$$

Verification:

$$\blacktriangleright \mathbf{A} \mathbf{x}_0 = (-3 + 4, -6 + 10, -9 + 14)' = (1, 4, 5)' = \mathbf{b}$$

$$\blacktriangleright \mathbf{A} \mathbf{v} = (7 - 6 - 1, 14 - 15 + 1, 21 - 21)' = 0$$

hence $\mathbf{A}(\mathbf{x}_0 + t \mathbf{v}) = \mathbf{b}$ for every $t \in \mathbb{R}$ ✓

Example: no solution (1/2)

We change only the last right-hand side of the previous system:

$$\begin{cases} 1 x_1 + 2 x_2 - 1 x_3 = 1 \\ 2 x_1 + 5 x_2 + 1 x_3 = 4 \\ 3 x_1 + 7 x_2 = 6 \end{cases} \quad (\mathbf{A}|\mathbf{b}) = \left(\begin{array}{ccc|c} 1 & 2 & -1 & 1 \\ 2 & 5 & 1 & 4 \\ 3 & 7 & 0 & 6 \end{array} \right)$$

With the same row operations $R_2 \leftarrow R_2 - 2 R_1$, $R_3 \leftarrow R_3 - 3 R_1$:

$$\left(\begin{array}{ccc|c} 1 & 2 & -1 & 1 \\ 0 & 1 & 3 & 2 \\ 0 & 1 & 3 & 3 \end{array} \right)$$

Example: no solution (2/2)

$$R_3 \leftarrow R_3 - R_2:$$

$$\left(\begin{array}{ccc|c} 1 & 2 & -1 & 1 \\ 0 & 1 & 3 & 2 \\ 0 & 0 & 0 & 1 \end{array} \right)$$

The last row corresponds to the equation $0 x_1 + 0 x_2 + 0 x_3 = 1$, which is impossible. Indeed $\mathbf{A}$ has 2 nonzero rows while $(\mathbf{A}|\mathbf{b})$ has 3:

$$\text{rank}(\mathbf{A}) = 2 < 3 = \text{rank}(\mathbf{A}|\mathbf{b}),$$

hence the system is inconsistent and has **no solution**.

Summary

Reading the ranks after forward elimination:

- ▶ Rank of a matrix in row echelon form = number of nonzero rows
- ▶ A row $(0 \ \cdots \ 0 \mid c)$ with $c \neq 0$: no solution
- ▶ Otherwise: pivot variables by backward substitution, $n - r$ free parameters

Parametric form:

- ▶ $\mathbf{x} = \mathbf{x}_0 + t \mathbf{v}$, with $\mathbf{A} \mathbf{x}_0 = \mathbf{b}$ and $\mathbf{A} \mathbf{v} = \mathbf{0}$

Homogeneous systems

Introduction: Homogeneous systems

- ▶ A homogeneous system is a system whose right-hand side is the zero vector
- ▶ Homogeneous systems always have at least one solution: we will see when they have other solutions
- ▶ They explain the structure $\mathbf{x} = \mathbf{x}_0 + t \mathbf{v}$ of the solutions of a general system, and they appear when computing eigenvectors

Homogeneous system

Definition 1: homogeneous system

*A system of linear equations $\mathbf{A} \mathbf{x} = \mathbf{b}$ with $\mathbf{A} \in \mathbb{R}^{m \times n}$ is called **homogeneous** if $\mathbf{b} = \mathbf{0}$, i.e., if it has the form*

$$\mathbf{A} \mathbf{x} = \mathbf{0}.$$

Solutions of a homogeneous system

Observation 1: solutions of a homogeneous system

Let $\mathbf{A} \in \mathbb{R}^{m \times n}$. Then:

- ▶ the homogeneous system $\mathbf{A} \mathbf{x} = 0$ always has the solution $\mathbf{x} = 0$, called the **trivial solution**;
- ▶ it has nontrivial solutions ($\mathbf{x} \neq 0$) if and only if $\text{rank}(\mathbf{A}) < n$; in this case it has infinitely many solutions, with $n - \text{rank}(\mathbf{A})$ free parameters;
- ▶ if $\mathbf{A} \in \mathbb{R}^{n \times n}$ is square, it has nontrivial solutions if and only if $\det(\mathbf{A}) = 0$.

Proof: Solutions of a homogeneous system

- ▶ Adding a column of zeros does not change the rank, hence

$$\text{rank}(\mathbf{A}|0) = \text{rank}(\mathbf{A})$$

and the system is always consistent (as shown by $\mathbf{x} = 0$).

- ▶ By the Rouché–Capelli theorem, the solution $\mathbf{x} = 0$ is the unique one if and only if $\text{rank}(\mathbf{A}) = n$; otherwise there are infinitely many solutions with $n - \text{rank}(\mathbf{A})$ free parameters.
- ▶ For a square matrix, $\text{rank}(\mathbf{A}) < n$ if and only if $\det(\mathbf{A}) = 0$.



Homogeneous systems: remarks

- ▶ In particular, if $m < n$ (fewer equations than variables), then $\text{rank}(\mathbf{A}) \leq m < n$ and the homogeneous system always has nontrivial solutions.
- ▶ If $\mathbf{x}_0$ is a solution of $\mathbf{A} \mathbf{x} = \mathbf{b}$ and $\mathbf{v}$ is a solution of $\mathbf{A} \mathbf{x} = 0$, then

$$\mathbf{A}(\mathbf{x}_0 + \mathbf{v}) = \mathbf{b} + 0 = \mathbf{b}.$$

This explains the parametric form $\mathbf{x} = \mathbf{x}_0 + t \mathbf{v}$ of the example with infinitely many solutions.

Example: a homogeneous system (1/2)

Consider the homogeneous system with the coefficient matrix of the example with infinitely many solutions:

$$\begin{cases} 1 x_1 + 2 x_2 - 1 x_3 = 0 \\ 2 x_1 + 5 x_2 + 1 x_3 = 0 \\ 3 x_1 + 7 x_2 = 0 \end{cases}$$

By the Sarrus rule,

$$\begin{aligned} \det(\mathbf{A}) &= 1 \cdot 5 \cdot 0 + 2 \cdot 1 \cdot 3 + (-1) \cdot 2 \cdot 7 \\ &\quad - (-1) \cdot 5 \cdot 3 - 2 \cdot 2 \cdot 0 - 1 \cdot 1 \cdot 7 \\ &= 0 + 6 - 14 + 15 - 0 - 7 = 0, \end{aligned}$$

so there are nontrivial solutions.

Example: a homogeneous system (2/2)

With the same row operations as in the example with infinitely many solutions (the last column stays equal to zero):

$$\left(\begin{array}{ccc|c} 1 & 2 & -1 & 0 \\ 0 & 1 & 3 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right)$$

Setting $x_3 = t$: $x_2 = -3t$ and $x_1 = -2x_2 + x_3 = 6t + t = 7t$.
Hence the solutions are

$$\mathbf{x} = t \begin{pmatrix} 7 \\ -3 \\ 1 \end{pmatrix}, \quad t \in \mathbb{R},$$

and the trivial solution is obtained for $t = 0$.

Summary

Homogeneous systems $\mathbf{A} \mathbf{x} = 0$:

- ▶ They always have the trivial solution $\mathbf{x} = 0$
- ▶ Nontrivial solutions $\iff \text{rank}(\mathbf{A}) < n$ (square case: $\det(\mathbf{A}) = 0$)
- ▶ If $m < n$, there are always nontrivial solutions

Structure of the solutions:

- ▶ Particular solution of $\mathbf{A} \mathbf{x} = \mathbf{b}$ + solution of $\mathbf{A} \mathbf{x} = 0$ = solution of $\mathbf{A} \mathbf{x} = \mathbf{b}$

LU Factorization Method



Introduction: LU Factorization Method

- ▶ The LU factorization writes the coefficient matrix as the product of a lower and an upper triangular matrix
- ▶ The original system is then split into two triangular systems, which are very easy to solve
- ▶ This is particularly convenient when we have to solve many systems with the same matrix **A** and different right-hand sides **b**

LU factorization

The **LU factorization** (or **LU decomposition**) is a method for solving systems of linear equations by decomposing the coefficient matrix **A** into the product of two triangular matrices:

$$\mathbf{A} = \mathbf{L} \mathbf{U}$$

where **L** is a **lower triangular matrix** (with 1's on the diagonal) and **U** is an **upper triangular matrix**.

Using the LU factorization

- ▶ Given the system of linear equations:

$$\mathbf{A} \mathbf{x} = \mathbf{b}$$

if we have the LU factorization $\mathbf{A} = \mathbf{L} \mathbf{U}$, we can substitute:

$$\mathbf{L} \mathbf{U} \mathbf{x} = \mathbf{b}$$

- ▶ We introduce the **auxiliary variable** $\mathbf{y}$ defined as:

$$\mathbf{y} := \mathbf{U} \mathbf{x}$$

so that $\mathbf{L} \mathbf{U} \mathbf{x} = \mathbf{b}$ becomes $\mathbf{L} \mathbf{y} = \mathbf{b}$.

Why does this work?

We are splitting the original system $\mathbf{A}\mathbf{x} = \mathbf{b}$ into two simpler triangular systems:

1. Find $\mathbf{y}$ such that $\mathbf{L}\mathbf{y} = \mathbf{b}$.
2. Find $\mathbf{x}$ such that $\mathbf{U}\mathbf{x} = \mathbf{y}$.

If both steps succeed, then

$$\mathbf{A}\mathbf{x} = \mathbf{L}\mathbf{U}\mathbf{x} = \mathbf{L}\mathbf{y} = \mathbf{b},$$

so $\mathbf{x}$ is indeed a solution of the original system.

Since both $\mathbf{L}$ and $\mathbf{U}$ are triangular, each of the two systems can be solved in $O(n^2)$ operations by simple substitution.

Step 1 - Forward substitution

Solve $\mathbf{L} \mathbf{y} = \mathbf{b}$ for $\mathbf{y}$

Since $\mathbf{L}$ is lower triangular, we can solve this system easily from top to bottom:

$$y_1 = b_1$$

$$y_2 = b_2 - \ell_{21} y_1$$

$$y_3 = b_3 - \ell_{31} y_1 - \ell_{32} y_2$$

$$\vdots$$

$$y_i = b_i - \sum_{j=1}^{i-1} \ell_{ij} y_j$$

Step 2 - Backward substitution

Solve $\mathbf{U} \mathbf{x} = \mathbf{y}$ for $\mathbf{x}$

Since $\mathbf{U}$ is upper triangular, we can solve this system easily from bottom to top:

$$x_n = \frac{y_n}{u_{nn}}$$

$$x_{n-1} = \frac{y_{n-1} - u_{n-1,n} x_n}{u_{n-1,n-1}}$$

$$\vdots$$

$$x_i = \frac{y_i - \sum_{j=i+1}^n u_{ij} x_j}{u_{ii}}$$

Advantages of LU factorization

- ▶ Once the factorization $\mathbf{A} = \mathbf{L} \mathbf{U}$ is computed, we can solve the system for different right-hand sides $\mathbf{b}$ efficiently
- ▶ Both forward and backward substitution require only $O(n^2)$ operations
- ▶ The factorization itself requires $O(n^3)$ operations, but it needs to be done only once

Example: LU factorization (1/5)

Consider the system of linear equations:

$$\begin{cases} 2x_1 + 1x_2 + 1x_3 = 4 \\ 4x_1 + 3x_2 + 3x_3 = 10 \\ 8x_1 + 7x_2 + 9x_3 = 24 \end{cases}$$

In matrix form: $\mathbf{A} \mathbf{x} = \mathbf{b}$ where

$$\mathbf{A} = \begin{pmatrix} 2 & 1 & 1 \\ 4 & 3 & 3 \\ 8 & 7 & 9 \end{pmatrix} \quad \mathbf{b} = \begin{pmatrix} 4 \\ 10 \\ 24 \end{pmatrix}$$

Example: LU factorization (2/5)

Given the LU factorization of $\mathbf{A}$:

$$\mathbf{L} = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 4 & 3 & 1 \end{pmatrix} \quad \mathbf{U} = \begin{pmatrix} 2 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \end{pmatrix}$$

Verification:

$$\mathbf{L} \mathbf{U} = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 4 & 3 & 1 \end{pmatrix} \begin{pmatrix} 2 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \end{pmatrix} = \begin{pmatrix} 2 & 1 & 1 \\ 4 & 3 & 3 \\ 8 & 7 & 9 \end{pmatrix} = \mathbf{A} \quad \checkmark$$

Example: LU factorization (3/5)

Step 1 - Forward substitution: Solve $\mathbf{L} \mathbf{y} = \mathbf{b}$

$$\begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 4 & 3 & 1 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = \begin{pmatrix} 4 \\ 10 \\ 24 \end{pmatrix}$$

From the first equation: $y_1 = 4$

From the second equation:

$$2 y_1 + y_2 = 10 \implies y_2 = 10 - 2 \cdot 4 = 10 - 8 = 2$$

From the third equation:

$$4 y_1 + 3 y_2 + y_3 = 24 \implies y_3 = 24 - 4 \cdot 4 - 3 \cdot 2 = 24 - 16 - 6 = 2$$

Example: LU factorization (4/5)

Thus: $\mathbf{y} = \begin{pmatrix} 4 \\ 2 \\ 2 \end{pmatrix}$

Step 2 - Backward substitution: Solve $\mathbf{U} \mathbf{x} = \mathbf{y}$

$$\begin{pmatrix} 2 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 4 \\ 2 \\ 2 \end{pmatrix}$$

From the third equation: $2 x_3 = 2 \implies x_3 = 1$

From the second equation: $x_2 + x_3 = 2 \implies x_2 = 2 - 1 = 1$

From the first equation:

$$2 x_1 + x_2 + x_3 = 4 \implies 2 x_1 = 4 - 1 - 1 = 2 \implies x_1 = 1$$

Example: LU factorization (5/5)

Solution:

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

Verification:

$$\mathbf{A} \mathbf{x} = \begin{pmatrix} 2 & 1 & 1 \\ 4 & 3 & 3 \\ 8 & 7 & 9 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 2+1+1 \\ 4+3+3 \\ 8+7+9 \end{pmatrix} = \begin{pmatrix} 4 \\ 10 \\ 24 \end{pmatrix} = \mathbf{b} \quad \checkmark$$

Summary

LU factorization $\mathbf{A} = \mathbf{L} \mathbf{U}$:

- ▶ $\mathbf{L}$ lower triangular with 1's on the diagonal, $\mathbf{U}$ upper triangular
- ▶ Solve $\mathbf{L} \mathbf{y} = \mathbf{b}$ by forward substitution, then $\mathbf{U} \mathbf{x} = \mathbf{y}$ by backward substitution

Cost:

- ▶ Factorization: $O(n^3)$ operations, done only once
- ▶ Each new right-hand side $\mathbf{b}$: only $O(n^2)$ operations

Cramer's rule

Introduction: Cramer's rule

- ▶ Cramer's rule gives an **explicit formula** for the solution of a square system with a unique solution, as a ratio of determinants
- ▶ We will see the formula for systems of two and three equations, with worked examples
- ▶ Finally, we compare its computational cost with Gaussian elimination: Cramer's rule is very useful for small systems, but not for large ones

Cramer's rule

Cramer's rule is an explicit formula for the solution of a system of m linear equations with m variables (valid whenever the system has a unique solution).

Cramer's rule: notation

- Given a column vector $\mathbf{b} \in \mathbb{R}^{m \times 1}$ of m rows, a matrix $\mathbf{A} \in \mathbb{R}^{m \times m}$ of m rows and m columns and a column vector $\mathbf{x} \in \mathbb{R}^{m \times 1}$ of m rows containing the m variables:

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1m} \\ a_{21} & a_{22} & \dots & a_{2m} \\ \vdots & \vdots & \dots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mm} \end{pmatrix} \quad \mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{pmatrix} \quad \mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_m \end{pmatrix}$$

Cramer's rule: formula

If $\det(\mathbf{A}) \neq 0$, the solution $\tilde{\mathbf{x}}$ of the system of m linear equations:

$$\mathbf{A} \mathbf{x} = \mathbf{b}$$

is given by the formula:

$$\tilde{x}_j = \frac{\det(\mathbf{A}_j)}{\det(\mathbf{A})}, \quad \forall j \in \{1, 2, \dots, m\}$$

where $\mathbf{A}_j$ is the matrix formed by replacing the j -th column of $\mathbf{A}$ by the column vector $\mathbf{b}$.

Systems of two equations and two variables (1/2)

With $m = 2$, we have:

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \quad \mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}$$

$$\mathbf{A}_1 = \begin{pmatrix} b_1 & a_{12} \\ b_2 & a_{22} \end{pmatrix} \quad \mathbf{A}_2 = \begin{pmatrix} a_{11} & b_1 \\ a_{21} & b_2 \end{pmatrix}$$

Systems of two equations and two variables (2/2)

The determinants are:

$$\det(\mathbf{A}) = a_{11} a_{22} - a_{12} a_{21}$$

$$\det(\mathbf{A}_1) = b_1 a_{22} - a_{12} b_2$$

$$\det(\mathbf{A}_2) = a_{11} b_2 - b_1 a_{21}$$

If $\det(\mathbf{A}) \neq 0$, we then have:

$$\begin{cases} a_{11} x_1 + a_{12} x_2 = b_1 \\ a_{21} x_1 + a_{22} x_2 = b_2 \end{cases}$$

$$\Rightarrow (\tilde{x}_1, \tilde{x}_2) = \left(\frac{\det(\mathbf{A}_1)}{\det(\mathbf{A})}, \frac{\det(\mathbf{A}_2)}{\det(\mathbf{A})} \right)$$

Example (1/4)

Given

$$\mathbf{A} = \begin{pmatrix} -1 & 1 \\ 8 & 2 \end{pmatrix} \quad \mathbf{b} = \begin{pmatrix} 2 \\ 19 \end{pmatrix}$$

we have

$$\mathbf{A}_1 = \begin{pmatrix} 2 & 1 \\ 19 & 2 \end{pmatrix} \quad \mathbf{A}_2 = \begin{pmatrix} -1 & 2 \\ 8 & 19 \end{pmatrix}$$

Example (2/4)

The determinants are:

$$\det(\mathbf{A}) = (-1) \cdot 2 - 1 \cdot 8 = -10$$

$$\det(\mathbf{A}_1) = 2 \cdot 2 - 1 \cdot 19 = -15$$

$$\det(\mathbf{A}_2) = (-1) \cdot 19 - 2 \cdot 8 = -35$$

Since $\det(\mathbf{A}) \neq 0$, we then have:

$$\begin{cases} -x_1 + x_2 = 2 \\ 8x_1 + 2x_2 = 19 \end{cases} \implies (\tilde{x}_1, \tilde{x}_2) = \begin{pmatrix} \frac{-15}{-10}, \frac{-35}{-10} \end{pmatrix} = \begin{pmatrix} \frac{3}{2}, \frac{7}{2} \end{pmatrix}$$

Example (3/4)

Given

$$\mathbf{A} = \begin{pmatrix} -1 & -1 \\ 1 & -1 \end{pmatrix} \quad \mathbf{b} = \begin{pmatrix} -2 \\ 0 \end{pmatrix}$$

we have

$$\mathbf{A}_1 = \begin{pmatrix} -2 & -1 \\ 0 & -1 \end{pmatrix} \quad \mathbf{A}_2 = \begin{pmatrix} -1 & -2 \\ 1 & 0 \end{pmatrix}$$

Example (4/4)

The determinants are:

$$\det(\mathbf{A}) = (-1) \cdot (-1) - (-1) \cdot 1 = 2$$

$$\det(\mathbf{A}_1) = (-2) \cdot (-1) - (-1) \cdot 0 = 2$$

$$\det(\mathbf{A}_2) = (-1) \cdot 0 - (-2) \cdot 1 = 2$$

Since $\det(\mathbf{A}) \neq 0$, we then have:

$$\begin{cases} -x_1 - x_2 = -2 \\ x_1 - x_2 = 0 \end{cases} \implies (\tilde{x}_1, \tilde{x}_2) = \left(\frac{2}{2}, \frac{2}{2} \right) = (1, 1)$$

Systems of three equations and three variables (1/3)

With $m = 3$, we have:

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \quad \mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$$

$$\mathbf{A}_1 = \begin{pmatrix} b_1 & a_{12} & a_{13} \\ b_2 & a_{22} & a_{23} \\ b_3 & a_{32} & a_{33} \end{pmatrix} \quad \mathbf{A}_2 = \begin{pmatrix} a_{11} & b_1 & a_{13} \\ a_{21} & b_2 & a_{23} \\ a_{31} & b_3 & a_{33} \end{pmatrix}$$

$$\mathbf{A}_3 = \begin{pmatrix} a_{11} & a_{12} & b_1 \\ a_{21} & a_{22} & b_2 \\ a_{31} & a_{32} & b_3 \end{pmatrix}$$

Systems of three equations and three variables (2/3)

and (Sarrus rule)

$$\begin{aligned}\det(\mathbf{A}) = & a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} \\ & - a_{13}a_{22}a_{31} - a_{12}a_{21}a_{33} - a_{11}a_{23}a_{32}\end{aligned}$$

$$\begin{aligned}\det(\mathbf{A}_1) = & b_1a_{22}a_{33} + a_{12}a_{23}b_3 + a_{13}b_2a_{32} \\ & - a_{13}a_{22}b_3 - a_{12}b_2a_{33} - b_1a_{23}a_{32}\end{aligned}$$

Systems of three equations and three variables (3/3)

$$\det(\mathbf{A}_2) = a_{11}b_2a_{33} + b_1a_{23}a_{31} + a_{13}a_{21}b_3 \\ - a_{13}b_2a_{31} - b_1a_{21}a_{33} - a_{11}a_{23}b_3$$

$$\det(\mathbf{A}_3) = a_{11}a_{22}b_3 + a_{12}b_2a_{31} + b_1a_{21}a_{32} \\ - b_1a_{22}a_{31} - a_{12}a_{21}b_3 - a_{11}b_2a_{32}$$

If $\det(\mathbf{A}) \neq 0$, we then have:

$$(\tilde{x}_1, \tilde{x}_2, \tilde{x}_3) = \left(\frac{\det(\mathbf{A}_1)}{\det(\mathbf{A})}, \frac{\det(\mathbf{A}_2)}{\det(\mathbf{A})}, \frac{\det(\mathbf{A}_3)}{\det(\mathbf{A})} \right)$$

Example (1/2)

We solve with Cramer's rule the system of the Gaussian elimination example:

$$\mathbf{A} = \begin{pmatrix} 2 & 1 & 1 \\ 4 & 3 & 3 \\ 8 & 7 & 9 \end{pmatrix} \quad \mathbf{b} = \begin{pmatrix} 4 \\ 10 \\ 24 \end{pmatrix}$$

We have

$$\mathbf{A}_1 = \begin{pmatrix} 4 & 1 & 1 \\ 10 & 3 & 3 \\ 24 & 7 & 9 \end{pmatrix} \quad \mathbf{A}_2 = \begin{pmatrix} 2 & 4 & 1 \\ 4 & 10 & 3 \\ 8 & 24 & 9 \end{pmatrix} \quad \mathbf{A}_3 = \begin{pmatrix} 2 & 1 & 4 \\ 4 & 3 & 10 \\ 8 & 7 & 24 \end{pmatrix}$$

Example (2/2)

The determinants are:

$$\det(\mathbf{A}) = 54 + 24 + 28 - 24 - 36 - 42 = 4$$

$$\det(\mathbf{A}_1) = 108 + 72 + 70 - 72 - 90 - 84 = 4$$

$$\det(\mathbf{A}_2) = 180 + 96 + 96 - 80 - 144 - 144 = 4$$

$$\det(\mathbf{A}_3) = 144 + 80 + 112 - 96 - 96 - 140 = 4$$

Since $\det(\mathbf{A}) \neq 0$, we then have:

$$(\tilde{x}_1, \tilde{x}_2, \tilde{x}_3) = \left(\frac{4}{4}, \frac{4}{4}, \frac{4}{4} \right) = (1, 1, 1),$$

which is the solution found with Gaussian elimination.

Computational cost (1/2)

- ▶ Cramer's rule gives an explicit formula, which is very useful for small systems ($m = 2$ or $m = 3$) and for theoretical purposes. However, it is **not** a practical method for large systems.
- ▶ To apply Cramer's rule we need $m + 1$ determinants of order m : $\det(\mathbf{A})$, $\det(\mathbf{A}_1)$, $\dots$, $\det(\mathbf{A}_m)$. If each determinant is computed with the Laplace expansion, the number of arithmetic operations grows roughly like $m!$ for each determinant.
- ▶ Gaussian elimination, instead, solves the system with a number of operations which grows roughly like m^3 (about $\frac{2}{3}m^3$ operations).

Computational cost (2/2)

- ▶ For example:
 - ▶ with $m = 10$ we have $10! = 3\,628\,800$, while $\frac{2}{3} \cdot 10^3 \approx 667$;
 - ▶ with $m = 20$, $20!$ is larger than $2 \cdot 10^{18}$, while $\frac{2}{3} \cdot 20^3 \approx 5\,333$.
- ▶ Even if the determinants are computed in a more efficient way (for instance with Gaussian elimination itself), Cramer's rule still needs $m + 1$ of them, and it remains more expensive than solving the system directly with Gaussian elimination.

Summary

Cramer's rule (square system, $\det(\mathbf{A}) \neq 0$):

- ▶ $\tilde{x}_j = \det(\mathbf{A}_j) / \det(\mathbf{A})$, where $\mathbf{A}_j$ is $\mathbf{A}$ with the j -th column replaced by $\mathbf{b}$
- ▶ For $m = 2$ and $m = 3$ the determinants are computed directly (Sarrus rule for $m = 3$)

Computational cost:

- ▶ Cramer's rule needs $m + 1$ determinants: useful for small systems only
- ▶ Gaussian elimination needs about $\frac{2}{3}m^3$ operations

Summary (1/2)

Existence and uniqueness:

- ▶ Rouché–Capelli: compare $\text{rank}(\mathbf{A})$, $\text{rank}(\mathbf{A}|\mathbf{b})$ and n
- ▶ No solution, unique solution, or infinitely many solutions with $n - r$ free parameters

Gaussian elimination:

- ▶ Forward elimination with elementary row operations, then backward substitution
- ▶ The row echelon form gives the ranks and the free parameters

Summary (2/2)

Homogeneous systems:

- ▶ Always the trivial solution; nontrivial solutions iff $\text{rank}(\mathbf{A}) < n$

LU factorization:

- ▶ $\mathbf{L} \mathbf{y} = \mathbf{b}$, then $\mathbf{U} \mathbf{x} = \mathbf{y}$: efficient for many right-hand sides

Cramer's rule:

- ▶ Explicit formula with determinants, useful only for small systems